TECHNICAL BRIEF · WEEK 4

Why We Must Automate Our Data Pipelines

Prepared by **Brian Nyamu**
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Every week, someone on our team manually re-types depot stock counts, sales numbers, and shipment records into a spreadsheet. It works — until it doesn't. Here's the risk we're carrying, the simple fix, and what it's worth to us.

1 The Risk: One Typo, One Bad Week

Picture this: a warehouse clerk closes out Friday's inventory count. Depot "C6" actually has **1,240** units of a product on the shelf, but the clerk mistypes it into the spreadsheet as **12,400** — one extra digit. Nobody catches it because nothing checks the number before it's used.

Monday morning, the replenishment plan sees a depot that looks over-stocked and **skips it** — while the real shelf is nearly empty. By midweek, C6 runs out of stock, deliveries are delayed, and a customer order is missed. One misplaced digit becomes a real financial and reputational cost — and it isn't a rare event. It's the kind of error that happens every time a human re-types a number by hand.

The uncomfortable truth: Excel doesn't know the difference between a real number and a typo. It will trust whatever we type — and pass that mistake straight into every decision built on top of it.

2 The Solution: Automatic Irrigation, Not Buckets

Think of our current process as watering a field by carrying buckets from a well. It works, but every trip depends on a person remembering to go, judging the right amount, and not spilling any along the way. An *automated pipeline* — a system that moves and checks data on its own, without anyone re-typing it — is like installing a sprinkler system instead: it draws from the same source, waters every row the same correct amount, on schedule, without anyone lifting a bucket.

TODAY — BUCKETS

Manual retyping, one field at a time. A tired hand, a fat-fingered key, or a missed row silently breaks the whole plan.

AUTOMATED — SPRINKLERS

Data flows straight from the source. Built-in checks (called *data validation*) catch typos and gaps automatically, before they reach a decision.

3 The Value: What It's Worth to Us

In this week's test run alone, the automated checks caught **over 4,300 inventory mismatches** and flagged **13,000+ instances** where a depot was at risk of running over capacity or out of stock — all before a single decision was made on bad numbers. A spreadsheet catches none of that on its own.

~90%	4,300+	Same-day
less time spent re-typing & reconciling data by hand	data errors caught automatically before reaching a report	stock-out warnings, instead of finding out a week later

Bottom line: Excel isn't wrong — it just can't check itself. Automation does.